Prathmesh Bele

PhD Student, National University of Singapore

prathmeshbele.github.io | prathmesh-bele | Scholar | prathmeshbele@u.nus.edu

Research Interests: Deep Learning, Computer Vision, Generative AI, Multimodal Learning

Education

Aug 2026 – Present | **National University of Singapore | School of Computing**
Doctor of Philosophy: Computer Science

Jul 2018 – Aug 2023 | **Indian Institute of Technology Bombay | Interdisciplinary Dual Degree** **Overall CPI: 8.99/10**
Master of Technology: Artificial Intelligence and Data Science
• *Achieved an CPI of **9.94/10** towards completion of Master of Technology*
Bachelor of Technology: Mechanical Engineering with ***Minor*** in Computer Science and Engineering

Research Experience

June 2024 – Present | **Google DeepMind** **Bangalore, India**
Pre-Doctoral Researcher | Advisors: Prof. Vinay Namboodiri, Dr. Vaibhav Rajan, Prof. Sara Beery
→ Leveraging sparse Street View via generative cross-view fusion to enhance satellite crop monitoring
→ Orchestrating curation of **1M+** spatiotemporal crop data points across India, aiming for global scale.
→ Impacting millions of farmers via core AI research enabling government optimizing resource allocation

Aug 2023 – Apr 2024 | **Mercedes-Benz Research and Development India** **Bangalore, India**
Post Graduate Engineer Trainee | Advisor: Dr. Laxmikant Sahoo
→ Harnessed complex multi-sensor data with the Perception team to drive scalable mapping solutions
→ Filed a **patent** for a novel GNN based approach achieving performance parity with production baselines

Dec 2021 – Apr 2023 | **IIT Bombay** **Mumbai, India**
Postgraduate Researcher | Advisors: Prof. Biplab Banerjee, Prof. Gemma Roig
→ Architected resilient models capable of generalizing from single sources under volatile domain shifts
→ Established novel invariant feature mechanisms, culminating in a top-tier publication [WACV'24]

Publications and Patents

[C] **Learning Class and Domain Augmentations for Single-Source Open-Domain Generalization** [🔗]
Prathmesh Bele, Valay Bundele, Avigyan Bhattacharya, Ankit Jha, Gemma Roig, and Biplab Banerjee
IEEE/CVF Winter Conference on Applications of Computer Vision **[WACV'24]**

[P] **Node and Edge Embedding Learning for Static Obstacle Segmentation**
Prathmesh Bele*, Apurva Kulkarni*, Hitul Desai, Amey Kavde, Dashrath Mali, Laxmikant Sahoo, et. al.
Intellectual Property of India, Reference No. 202341087789 **[IP India]**

* denotes equal contributions, C = Conference Publication, P = Patent

Selected Research Projects

Multi-Sensor Cross-View Panoramic Synthesis | Google DeepMind
Advisors: Prof. Vinay Namboodiri, Dr. Vaibhav Rajan
→ Built a diffusion model synthesizing **physically consistent** panoramas from solely low-resolution satellite data.
→ Internalized HR structural priors via a **cyclic critic**, eliminating explicit geometry needs at inference.
→ Bridged polar aerial and spherical ground geometries using a **coordinate-aware cross-attention** mechanism.
→ Outperformed explicitly HR-conditioned baselines across all metrics on a global dataset of **550K** images.

Cross-View Generative Fusion | Google DeepMind
Advisors: Dr. Vaibhav Rajan, Prof. Sara Beery
→ Pioneering unified **cross-view generative fusion** to overcome fine-grained detail loss in decoupled methods.
→ Bridged a critical data gap by curating a **novel dataset** temporally aligning Sentinel-2 with agricultural Street View
→ Architecting **discrete diffusion** models to resolve fundamental continuous-discrete latent incompatibilities

Single Source Open Domain Generalization | IIT Bombay
Advisors: Prof. Biplab Banerjee, Prof. Gemma Roig

→ Developed **SODG-Net**, an end-to-end network outperforming benchmarks by **1-14%** in open-domain settings
→ Proposed a style synthesis block generating diverse statistical features to effectively simulate novel domains
→ Formulated novel weight learning and margin objectives to establish distinct representations for open classes

Graph Neural Networks for Free-space Mapping | MBRDI
Advisor: Dr. Laxmikant Sahoo

→ Proposed a novel **GNN** architecture formulating free-space mapping as a scalable edge-prediction problem [Patent]
→ Architected a dual-path **Graph Attention Network** to independently process node and edge features
→ Implemented class-aware edge retention via feature fusion, achieving parity with inflexible rule-based legacy systems

Crop Classification & Data Curation | Google DeepMind
Advisor: Alok Talekar

→ Leveraged **Gemini** to semantically filter Street View imagery, isolating optimal candidates for expert annotation.
→ Established comprehensive annotation guidelines to curate a high-quality, ground-level crop dataset.
→ Developed a pipeline augmenting scarce satellite data with robust, timestamped, Street View-derived crop labels.

Incremental Domain Adaptation | IIT Bombay
Advisors: Prof. Biplab Banerjee, Prof. Gemma Roig

→ Devised a robust solution establishing **discriminative, domain-invariant** feature spaces under shifting distributions.
→ Retained critical prior learnings by aligning model outputs and batch-norm parameters for representative proxy data.
→ Leveraged center loss alongside standard cross-entropy to significantly enhance model prediction confidence.

Honors and Achievements

→ Among **15**, from **18,000+** applicants, selected for Google's Pre-Doctoral Researcher Program 2024
→ Selected among the Top **3000** students nationwide for the exclusive **Amazon Machine Learning Summer School** 2022
→ Awarded **INSPIRE Scholarship** for ranking in the **Top 1%** of the Maharashtra Board Class 12th Examinations 2017
→ Secured **99.32 Percentile** in JEE Main (**1.5M** entrants) and **98.75 Percentile** in JEE Advanced (**0.2M** entrants) 2018
→ Recipient of scholarships for securing **1st Rank** in District-level Maharashtra Talent Search Examination 2015

Notable Positions of Responsibility

→ **Reviewer**: **CVPR 2026, ECCV 2026, NeurIPS 2026, SPoT Workshop (ICLR) 2026** Present
→ **Teaching Assistant, IITB**: Managed tutorials & grading for **150+ students** in ML and Data Science courses 2019-21

Exploratory Research

→ **HSI Unmixing:** Developed Vision Transformer for HSI unmixing with adaptive skips connections, with Prof. B. Banerjee
→ **RSVQA:** Built modified DenseNet with attention decoder for Remote Sensing VQA (76.63% acc.), with Prof. S. Sarawagi
→ **Binarized NN:** Implemented Binarized NN via Info Bottleneck for object detection, with Prof. P. Balamurugan
→ **Multimodal Fusion:** Fused LiDAR-HSI via attention for land cover classification (80% acc.), with Prof. B. Banerjee
→ **3D MeshNet:** Constructed MeshNet for 3D object recognition from triangular meshes, Seasons of Code, IITB
→ **Taxi Demand:** Predicted NYC taxi demand/duration using LSTM and ARIMA time-series models, with Prof. A. Sethi
→ **Stock Prediction:** Analyzed stock trends using K-Means and LSTM for time-series forecasting, with Prof. A. Tewari
→ **Texture Synthesis:** Implemented patch-based image quilting for texture synthesis and transfer, with Prof. S Awate

Key Courses Undertaken

Machine Learning Advanced Machine Learning, Automatic Speech Recognition, Fundamentals Intelligent and Learning Agents, Fundamentals of Digital Image Processing, Statistical Machine Learning & Data Mining, Machine Learning for Remote Sensing, Deep Learning - Theory and Practice, Introduction to Machine Learning

CS and Maths Design & Analysis of Algorithms, Operating Systems, Data Structures & Algorithms, Programming for Data Science, Linear Algebra, Calculus, Differential Equations, Numerical Analysis

Technical Skills

Programming Languages Python, C++
Frameworks/Tools PyTorch, TensorFlow, Jax/Flax, MATLAB, Git, LaTeX